

CRITERIA TO EVALUATE NEUROGENIC BOWEL IN CHILDREN WITH CONGENITAL ZIKA SYNDROME

Authors: Valéria Azevedo de Almeida¹, Nancy Sotero Silva¹, Lilian Lira Lisboa¹, Rafael Pauletti Gonçalves¹, Edgard Morya¹, Lucia Maria Costa Monteiro², Reginaldo Antônio de Oliveira Freitas Júnior¹.

¹ Institution: Santos Dumont Institute of Education and Research (ISD), Macaíba, Brasil.

² National Institute of Women, Children and Adolescents Health Fernandes Figueira (IFF/FIOCRUZ), Rio de Janeiro, Brasil.

Protocol recommendation: refer all children with congenital zika syndrome to bowel function assessment, according to the following:

1. Initial Assessment:

- a) It must contain detailed clinical history, describing bowel habits, including continence, water intake and diet type, current medications and any special regimen for bowel emptying, as well as frequency of bowel emptying and stool characteristics.

2. Bristol Stool Scale:

- a) It is recommended the use of Bristol Stool Scale, which consist of a measure of visual classification that helps patients reporting feces consistency. It aims to evaluate, from descriptive ways, fecal content shapes using pictures which represent seven types of feces, according to its shapes and consistency (MARTINEZ, 2012).

3. Rome IV Criteria

- a) To assess gastrointestinal function it is recommended the use of Rome IV criteria, which are the most updated guideline for diagnosis of functional constipation in children. Reduction from two to one month of symptoms duration, regarding Rome III criteria, search for harmonize with guideline developed in 2014. Rome Criteria are widely used to standardize diagnosis of constipation, mainly at researches.

b) For children from 0 to 4 years old, it is recommended the following criteria:

- Two or fewer defecations per week;
- History of excessive stool retention;
- Retentive posturing;
- History of hard or painful bowel movements;
- Presence of fecal mass in the rectum.

4. Ultrasound:

a) It is recommended the measurement of Rectal Diameter (RD) by abdominal ultrasound for evaluation of constipation and fecal impaction.

b) $RD \geq 2,9$ it's compatible with bowel constipation, as evidenced by JOENSSON *et al.* (2008) and BURGERS *et al.* (2013).

Follow-up:

- a) Parents must be oriented as for the realization of behavioral treatment for evacuation.
- b) Clinical reevaluation each two or three months, depending on the response to treatment.

References

BURGERS, R.; DE JONG, T. PVM.; BENNINGA, M. A. Rectal examination in children: digital versus transabdominal ultrasound. **The Journal of urology**, v. 190, n. 2, p. 667-672, 2013.

DROSSMAN, D. A.; HASLER, W. L. Rome IV—functional GI disorders: disorders of gut-brain interaction. **Gastroenterology**, v. 150, n. 6, p. 1257-1261, 2016.

JOENSSON, I. M. *et al.* Transabdominal Ultrasound of Rectum as a Diagnostic Tool in Childhood Constipation. **Journal of Urology**, v. 179, n. 5, p. 1997–2002, 2008.

LEVY, E. I.; LEMMENS, R.; VANDENPLAS, Y.; DEVREKER, T. Functional constipation in children: challenges and solutions. **Pediatric health, medicine and therapeutics**, Dove Press, v. 8, p. 19, 2017.

MARTINEZ, A. P.; DE AZEVEDO, G. R. Tradução, adaptação cultural e validação da Bristol Stool Form Scale para a população brasileira. **Revista Latino-Americana de Enfermagem**, v. 20, n. 3, p. 583-589, 2012.